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Masculine and Feminine Writing in Atomic Bomb Literature: A Sentiment Analysis Approach

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Overview



1. Japanese Atomic Bomb Literature
2. Feminine Sentiment Protection: Hypothesis Statement
3. Sentiment Analysis for Japanese: Oseti and Transformers-models
4. Sentiment Difference Distribution
5. Conclusions



Japanese Atomic Bomb Literature



Atomic Bomb Literature (*genbaku bungaku*) is a direction in Japanese post-WWII literature which explored the experiences and aftermath of the atomic bombings of Hiroshima and Nagasaki in August 1945. This genre depicts the human suffering, moral dilemmas, and societal impacts caused by the atomic bombings. Also, the genre can be considered as a part of Japanese War Literature.





Feminine Patterns in Women's War Literature - Informal Statement



Informal statement:

Women's writings, excluded from the masculine, "combatant gnostic" canon of war literature, found the "homefront" as women's genuine territory and make women in this manless wartime environment major characters. Experiencing wartime non-combatant trauma, they imply counter-traumatic care towards their characters. In particular, they separate their female protagonists from wartime negativity, creating isolated from war narrative spaces, and take burden of expressing inevitable war negativity on their authorial voices.



Feminine Patterns in Women's War Literature - Formal Statement



Formal statement:

Female writers isolate their characters from war negativity by making direct speech of their characters having a higher sentiment score than a sentiment score of author's speech. On the contrary, male writers do not demonstrate this feature.

Sentiment difference (SD) = average direct speech sentiment - average author's speech sentiment

Female writers: average (median) of SDs > 0 ?

Male writers: average (median) of SDs > 0 ?

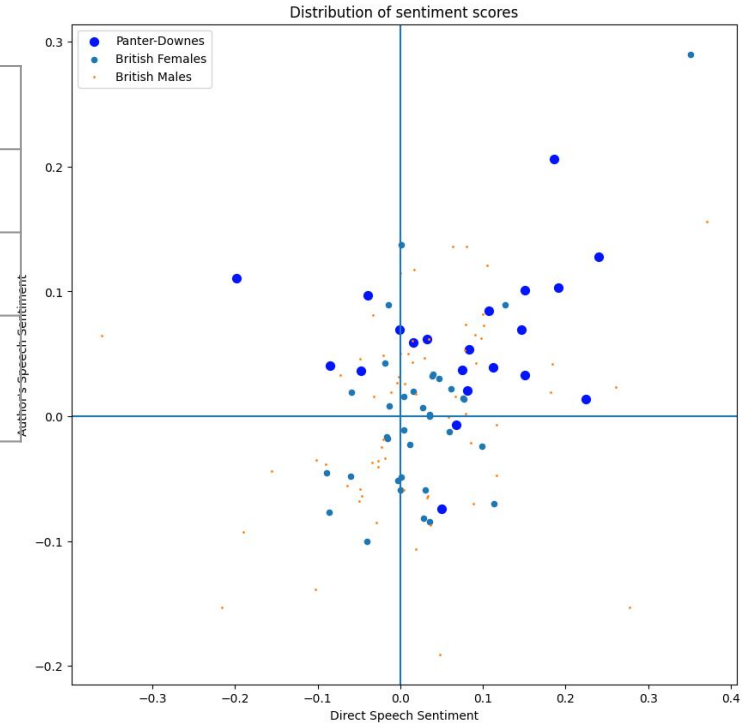
Previous Case - British WWII Literature



Brief characteristics:

	Female	Male
No. of writers	20	22
No. of pieces	56	66
SD (median)	0.15	-0.34

Result: clear distinction between two sets of writers by the sentiment difference





Corpus Description



Source: *Nihon no genbaku bungaku*
(*Japanese Atomic Bomb Literature*) in
15 volumes;

177 fiction fragments (76 by male and
41 by female);

42 authors (10 female and 32 male)

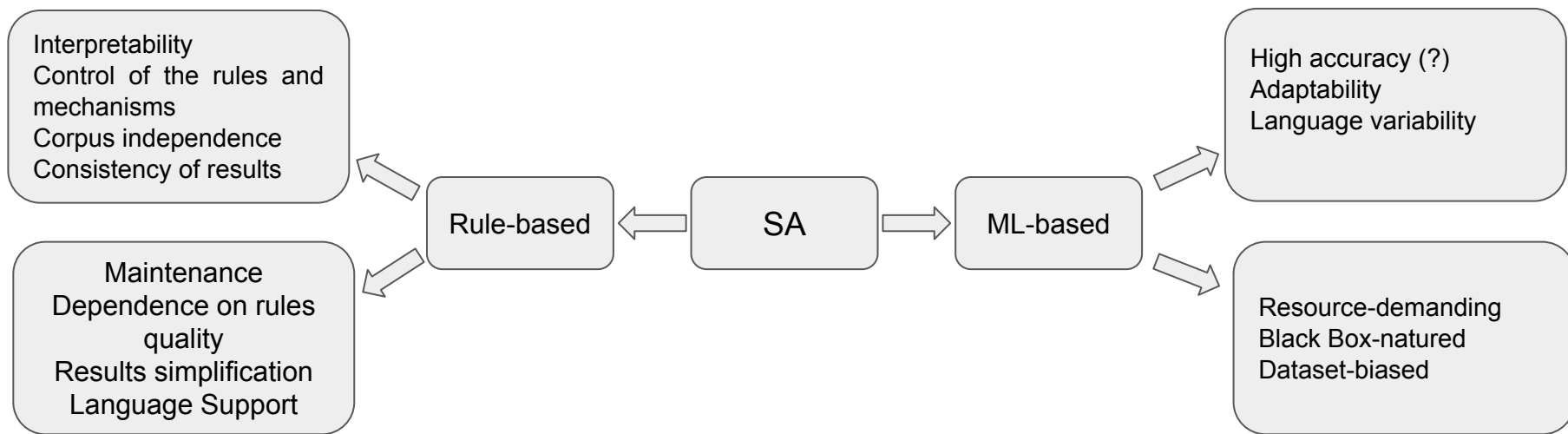
Corpus processing:

OCRing with ABBYY FineReader and
post-processing of common
OCR-errors.





Sentiment Analysis (SA) is a broad term for instruments that analyze “people's opinions, sentiments, appraisals, attitudes, and emotions towards entities and their attributes (Liu 2020).”





Oseti & VADER Characteristics



VADER (Valence Aware Dictionary and sEntiment Reasoner)

by (Hutto & Gilbert 2014)

Lexicon and Rule-based

Language: English

provides four scores:

Positive: Proportion of text that is positive.

Negative: Proportion of text that is negative.

Neutral: Proportion of text that is neutral.

Compound: A normalized, weighted composite score that ranges from -1 (extremely negative) to +1 (extremely positive).

Oseti

by (Ikegami 2021)

Lexicon-based (considers negation)

Language: Japanese

Source of lexicon: 日本語評価極性辞書 (Dictionary of Japanese Language Evaluation)

Outputs the score $[-1, +1]$ which is calculated as weighted difference between a number of negative- and positive-marked words in the sentence.

Oseti vs VADER Juxtaposition: Random Sampling



Sampling: 159 random sentences from the corpus and 159 Google-translated sentences

	Oseti	VADER
Sentiment scores (mean)	0.06	0.002
Sentiment scores (std)	0.53	0.38
Zero scores in Oseti for non-zero in VADER (number/share)	30 / 0.19	
Oseti/Vader scores correlation	0.4	
Oseti matching polarity with Vader, excluding 30 zero scores (number/share)	38/0.29	



Oseti vs VADER Juxtaposition: Direct Speech Sampling



Sampling: 161 random direct speech segments from the corpus and 161 Google-translated direct speech segments

	Oseti	VADER
Sentiment scores (mean)	-0.12	-0.12
Sentiment scores (std)	0.39	0.60
Zero scores in Oseti for non-zero in VADER (number/share)	25 / 0.16	
Oseti/Vader scores correlation	0.28	
Oseti matching polarity with VADER, excluding 25 zero scores (number/share)	77/0.57	



Why not transformers?



21 models on Hugging Face

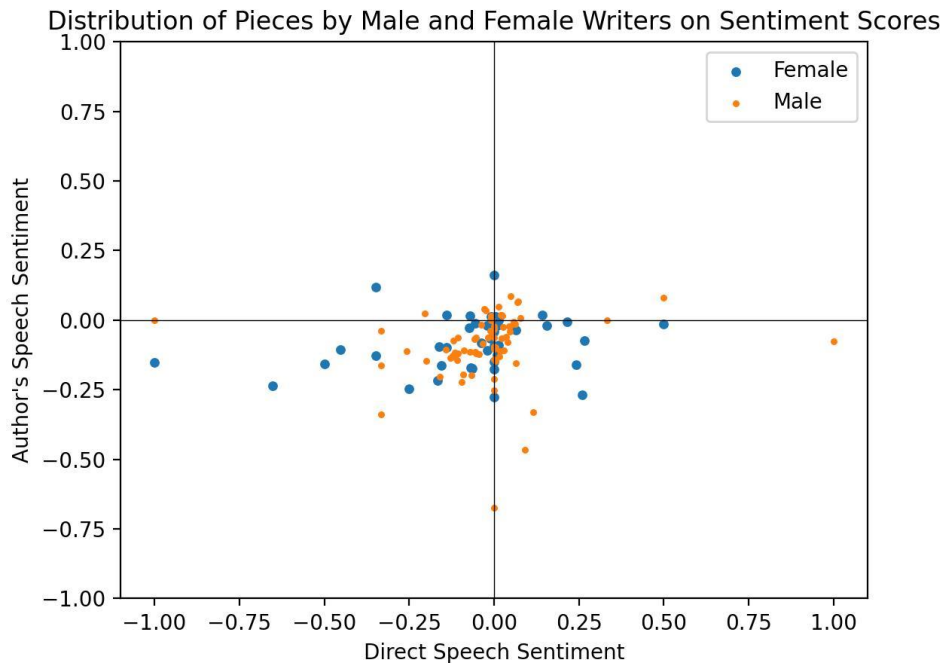
Unappropriate training datasets both by the size and thematic field:

- bert-finetuned-japanese-sentiment (Phu 2023): 20000 Amazon review
- japanese-sentiment-analysis(Patel 2022): 220 corporate financial reports (6,119 sentences)
- Finance-sentiment-ja-base (bards.ai 2023): Japanese financial news (about 5,000 sentences/phrases)
- Japanese Stock Comment Sentiment Model (c299m 2023): comments and discussions related to Japanese stocks (dataset not described)
- Others - undocumented

Atomic Bomb Literature Sentiment Analysis - 1



	Female writers	Male writers
SD (mean)	0.015	0.063
SD (median)	0.041	0.040
Negative SD cases (number/ratio)	17/0.41	19/0.25

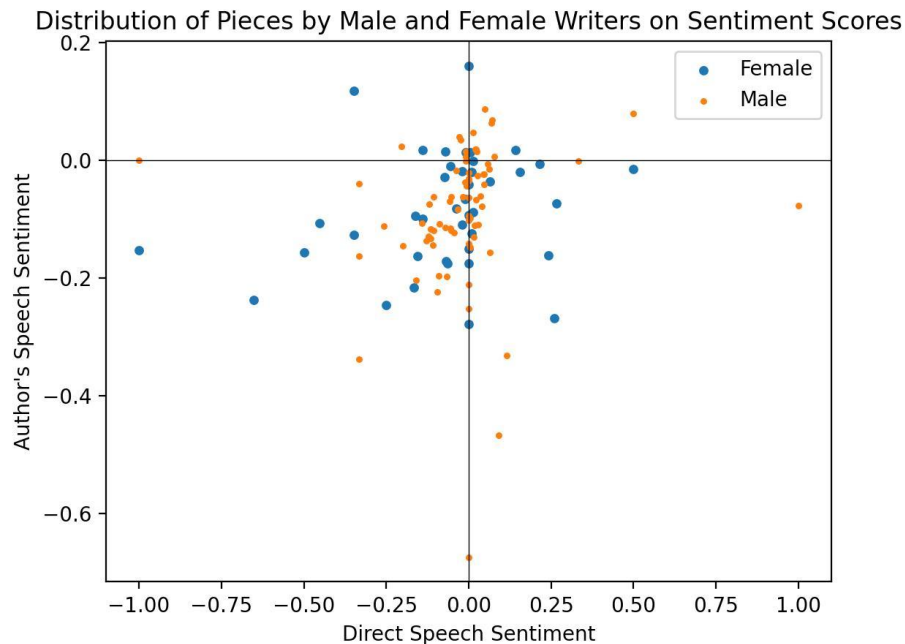




Atomic Bomb Literature Sentiment Analysis - 2



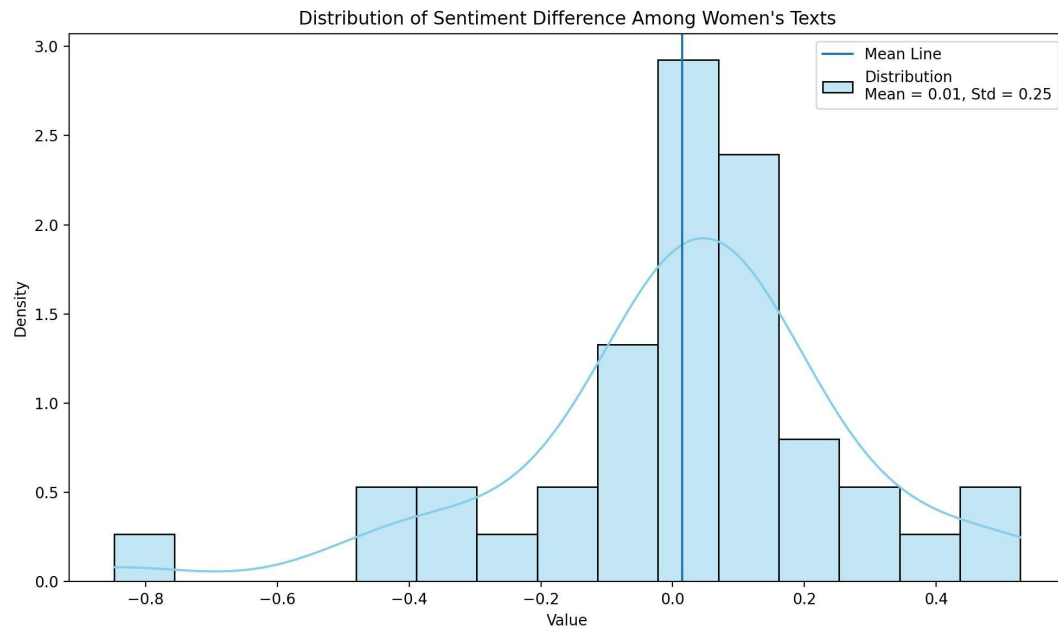
	Female writers	Male writers
SD (mean)	0.015	0.063
SD (median)	0.041	0.040
Negative SD cases (number/ratio)	17/0.41	19/0.25



Sentiment Difference Distribution in Women's Texts



The SD scores have a disproportionate distribution, and the major contribution is made by negative outlier cases.





Outlier Negative Sentiment Difference Works: Female Works



Original author and Title	Translated	Length in characters	Overall Sentiment	Sentiment Difference
大田 洋子 - 屍の街序(冬芽書房版)	Yoko Ota - Introduction to the City of Corpses (Toume Shobo edition)	3644	-0,21237	-0,84782
大田 洋子 - 原子爆弾	Yoko Ota - Atomic Bomb	2173	-0,2942	-0,4651
大田 洋子 - たましの後遺症	Yoko Ota - Aftereffects of the soul	2866	-0,31296	-0,41536
大田 洋子 - 私と「原爆症」とについて	Yoko Ota - About me and "Atomic Bomb Disease"	6037	-0,15759	-0,3456
大田 洋子 - ヒカドン はごめんだ	Yoko Ota - Hikadon is sorry!	6660	-0,16753	-0,34358
大田 洋子 - 私は忘れない広島思い出を	Yoko Ota - Memories of Hiroshima that I want to forget	7517	-0,15443	-0,22036

Ota Yoko is the most “pessimist” Atomic Bomb Literature writer.

Outlier Positive Sentiment Difference Works: Female Works



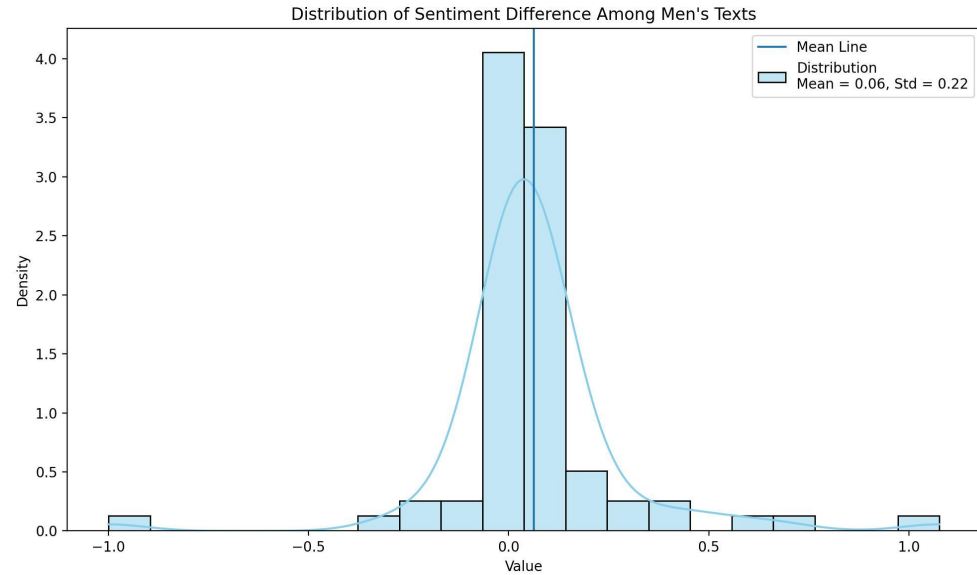
Original author and Title	Translated	Length in characters	Overall Sentiment	Sentiment Difference
大田 洋子 - 創作態度	Yoko Ota - Creative attitude	1548	-0,08492	0,339796
美川 きよ - あの日のこと	Kiyo Mikawa - About that day	6988	-0,12386	0,401536
林 京子 - 昭和二十年の夏	Kyoko Hayashi - Summer of 1945	18826	-0,01325	0,51509
大田 洋子 - 広島の木の子	Yoko Ota - Hiroshima Tree Bark	3725	-0,24347	0,527981

However, the most extreme cases of feminine protection also belong to Ota Yoko (and Hayashi Kyoko and Mikawa Kiyo).

Sentiment Difference Distribution in Men's Texts



Texts, written by men, tend to more even distribution of SD. Unlike female writers, all of them tend to follow a particular canonical path of expressing the relations between characters protection and negativity of author's narrative voice.



Outlier Negative Sentiment Difference Works: Male Works



Original author and Title	Translated	Length in character s	Overall Sentiment	Sentiment Difference
大江 健三郎 - ヒロシマ・ノート 4	Kenzaburo Oe - Hiroshima Notes 4	14323	-0,00013	-0,99948
中井 正文 - 名前のない男	Masafumi Nakai - Anonymous Man	25890	-0,04393	-0,29397
大江 健三郎 - 核状況のカナリア理論	Kenzaburo Oe - Canary Theory of Nuclear Situation	16313	0,006504	-0,22867
井上 光晴 - 病み犬を連れた老婆 『地の群れ』の舞台	Mitsuharu Inoue - An old woman with a sick dog, the setting for "The Crowd of the Earth"	2709	-0,16312	-0,17021
大江 健三郎 - ヒロシマ・ノート 7	Kenzaburo Oe - Hiroshima Notes 7	10756	-0,11935	-0,14579

Oe Kenzaburo's *Hiroshima Notes* (part 4) is the only extreme case of the masculine negation of sentiment protection, while other remarkable negative SD cases vary within about 1 σ .

Outlier Positive Sentiment Difference Works: Male Works



Original author and Title	Translated	Length in characters	Overall Sentiment	Sentiment Difference
金井利博 - 核権力ヒロシマの告発6	Toshihiro Kanai - Accusation of Nuclear Power Hiroshima 6	482	0,113333	0,42
井上 光晴 - 生者も死者も被爆者 原点への引戻し	Mitsuharu Inoue - A-bomb survivors both living and dead - Returning to the origin	2217	-0,14037	0,446496
井上 光晴 - 「前科三犯」の被爆者	Mitsuharu Inoue - A-bomb survivor with three previous convictions	652	-0,2	0,557576
原 民喜 - 原爆被災時のノート	Tamiki Hara - Notes from the time of the atomic bombing	2901	-0,67395	0,673947
梶山 季之 - 実験都市	Toshiyuki Kajiyama - Experimental City	20182	-0,07676	1,076533

Kajiyama's *Experimental City* is a true outlier as it is less related to *genbaku bungaku*, but touches the problem of rapid technological progress and its consequences, including Atomic Bomb.



Conclusions



- In general, it was not proven that female writers share feminine mode of writing in *genbaku bungaku*.
- The negative SD in women's subcorpus is defined more by negative outlier cases, presented by Ota Yoko's texts, while the number of negatively sentimented cases reveals less fitting into feminine expectations.
- Ota Yoko also demonstrates a variety of feminine and masculine SD modes in her writings.
- Men's writings are less variative and follow genre's canonical line, demonstrating a more protective SD mode.
- This suggests a nuanced landscape where gender itself does not strictly dictate narrative sentiment strategies but shows that the genre is gendered itself and contains a more widespread "feminine" modus, surprisingly more popular among male writers, and a "masculine" one, which does not share that sentiment narrative caregiving strategy, which is more evident in texts written by female authors.



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